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Effect of nitrogen atmosphere and vacuum conditions on charge dynamics in mesoscale porous silicon

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ABSTRACT

Optoelectronic properties of nano- and mesoscale porous silicon (PS) are strongly influenced by surface effects, owing to intrinsic and extrinsic electronic states within the bandgap. In addition, physisorption, chemisorption, and desorption of environmental species further alter the surface electronic structure, thereby modifying the surface potential barrier. However, the influence of gaseous environments and vacuum conditions on surface charge dynamics in PS is not fully understood. In this work, we systematically investigated environment-dependent surface charge dynamics in PS using transient surface photovoltage (SPV)—a non-destructive, highly surface-sensitive characterization technique. Our measurements revealed a multi-component photoresponse sensitive to ambient conditions, indicating physisorption/chemisorption-mediated charge exchange processes. Comparison of results obtained in gaseous environments and in high vacuum enabled separation of contributions to band bending from intrinsic surface states and from the adsorbate-induced ones. Furthermore, the study of the influence of illumination intensity on SPV transients showed that their onset slopes exhibit nonlinear scaling with illumination intensity, in contrast to the linear behavior typically observed in bulk semiconductors, suggesting a drift/diffusion-mediated charge redistribution in PS. By revealing environment-dependent transport behavior, this study provides insight into illumination-governed carrier dynamics as well as guidelines for the rational design of PS-based sensing devices.

I. INTRODUCTION

Silicon is the most common material used in modern electronic devices, and the growing demand for compact devices is driving the miniaturization of silicon-based modules. Although the bulk crystalline silicon is of exceptional value to electronics, its applications in the optical, magnetic, or biophysical fields remain limited. However, when the size of silicon structures is reduced to the nanometer scale, its physicochemical properties undergo significant changes, which opens up broad prospects for use in nanotechnology. Due to its photoluminescence capacity and high biocompatibility, nanostructured silicon is becoming a promising material for optoelectronic and biomedical applications.[1,2]

Nano- and mesoscale porous silicon (PS) has been gaining remarkable attention as one of the promising materials for optoelectronic applications in devices such as light emitters,[3] photodetectors,[4] photovoltaic cells,[5] etc., due to porous silicon's wide bandgap,[6,7] high luminescence efficiency,[8,9] and compatibility with Si-based technologies.[10] Its large internal surface area and reactivity with environmental species open further possibilities for sensing[11] and biomedical applications.[12] The tunable morphology renders it a versatile host for filling with various materials such as polymers[13] or magnetic materials.[14,15] A detailed discussion of the surface chemistry and stabilization of PS, which are critical for these applications, is provided in Ref. 16. To further understand the formation and morphology of PS, computer simulations have been carried out for various doping levels of the crystalline substrate, temperatures, HF concentrations, and anode current densities, which show that porosity depends on the depth and reveals the presence of a fractal dimensionality on scales of less than 10 nm.[17,18] Chemical, physical, and electrical processes involved in the PS growth mechanism are investigated numerically by varying the morphology parameters in two dimensions.[19] Moreover, while PS yields controlled and scalable morphologies, a number of reports on its electrical and photo-electrochemical properties have shown that PS can be very conductive and photoactive,[20,21] essential for serving as a base material for optoelectronic gas sensors. In this regard, one of the most important issues is elucidation of illumination-induced charge recombination processes in different environmental conditions. A large variety of extrinsic and intrinsic states exist at both the SiO_2 surface and at the interface between Si and SiO_2. When the surface of PS is illuminated, the charge transfer could occur via many of these states, and the rate of this transfer may occur on different time scales depending on the specific recombination channel. For example, the surface-adsorbed species could take a longer time as the bulk charge overcomes the high near-surface barrier. Moreover, the pressure of the environment could greatly affect the interaction between adsorbed species and PS, which, in turn, can alter the rate of near-surface charge recombination. Thus, in order to tailor the performance of PS-based optoelectronic gas sensors, it is important to determine response and recovery times under illumination and in the dark as well as other figures of merit such as reversibility and selectivity under different pressures.

The photon energy of the incoming radiation directly affects the charge recombination and hence the response times. A sub-bandgap illumination may directly affect the optical cross sections of surface states, whereas a super-bandgap illumination may affect thermal cross section resulting in different response times.

Thus, for PS to find applicability for optoelectronic gas sensors, knowledge of the most suitable spectral range(s) is of great interest. Naturally, the surface photoresponse could be affected not only by the excitation wavelength but also by the intensity of the incident illumination. By observing the rate at which the surface potential responds to changes in light intensity, one can gain insight of the relative contributions of drift- and diffusion-related processes of the photon-induced electronic transport.

Some of the issues related to the environments such as the effect of a gas,[22] organic vapors,[23] humidity[24] on the electronic properties of PS have been investigated in a number of studies. Only a few works have addressed the effect of the surface band bending on PS under illumination separately either in gas or vacuum. Surface photovoltage (SPV) measurements on PS on a substrate and in a free form were carried out only in vacuum using a laser to characterize processes occurring on rather short time scales (within a few seconds).[25,26] In general, the charge recombination due to the illumination of the surface covered with adsorbates may not be trivial and response/recovery may take hours or days to saturate, though it has been overlooked.

Most of the time irradiation by either poly- or monochromatic light is used to monitor the change in surface potential of different materials including PS.[27] The rate of change of the surface barrier immediately after the onset of irradiation for most of the bulk semiconductors was found to be linearly proportional to the illumination intensity, which is very useful to find different surface-related parameters, such as initial band bending, optical cross section, and thermal capture cross section.[28–31]

Apart from those parameters, other optoelectronic properties of PS with partially oxidized surfaces have not been studied in detail. Such a study would be justified for better understanding of the photophysical processes occurring in PS structures as well as for the development of optoelectronic gas sensors. To the best of our knowledge, no comparative work has been done to address the effect of ambient pressure on the illuminated surface of PS. Studies of surface transport properties under different ambient pressures could allow better understanding of the separate roles played in the photoinduced charge exchange by the Si/SiO_2 interface and PS-free surface. Moreover, one of the most important figures of merit related to sensors, the reversibility, has not been explored in depth.

In order to study reversibility, one has to understand the reaction of PS during illumination and in the dark under different environmental conditions. From the previous works on bulk semiconductors, the recovery time is always slower than the response time; however, it is not clear whether this is valid for nanoscale materials such as PS. So far, the search for suitable materials for pressure sensing with high adsorption/desorption kinetics has resulted in limited success. Only very recently, Aouass *et al.*[32] reported the creation of a defect-free surface through the desorption of native oxide and the suppression of pore formation on the PS surface. This was achieved via *in situ* sintering in an ultrahigh-vacuum chemical vapor deposition reactor under a hydrogen environment, followed by heating at 900 °C. Building on this work, another report further demonstrated that annealing a PS substrate with epitaxially grown SiGe improves surface morphology of SiGe through stress relaxation and internal microstructure stabilization. Notably, annealing at 1100 °C yielded higher crystalline quality compared to unannealed PS substrates or those annealed at 900 °C or below.[33] Therefore, PS could be a low-cost sensing material with strong potentials of pressure selective and reversible properties, which can operate at room temperature. However, no studies were performed in different ambients to make a comparison between different excitations for the same material, and PS is no exception in this regard. Since the response time during the irradiation by polychromatic and/or monochromatic light may show single or multiple relaxation times, it is, thus, crucial to identify whether a specific excitation wavelength is responsible for responses on multiple time scales. Similarly, the behavior of onset slope vs illumination intensity in the case of a material, such as PS, has not been adequately studied. Such measurements could be a very efficient way to elucidate the effect of multiple processes defining the transport of non-equilibrium charge carriers in nanoscale materials.

In this work, we address all these issues employing PS samples obtained by anodizing a silicon substrate.[34] This PS is a versatile material that can exhibit different morphologies depending on the level of doping of the substrate and other growth parameters. We present a systematic approach to investigate illumination-induced charge separation in PS under different

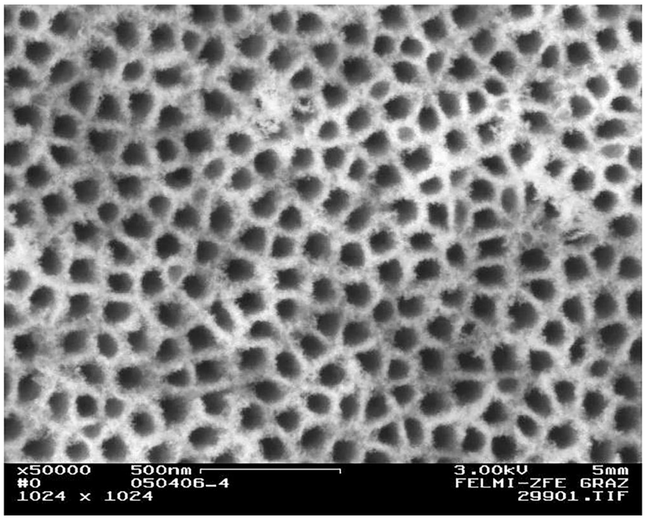

FIG. 1. SEM image of the top view of the PS sample with a pore diameter of ∼60 nm, distance between pores of ∼45 nm, and thickness of porous layer of ∼35 μm.

ambient pressures and discuss its sensitivity and reversibility using a highly surface-sensitive probe, transient and spectral surface photovoltage (SPV), which is largely insensitive to the environment and works in a broad range of energies and times. We also investigate the effect of incident wavelengths on charge dynamics and intensities on initial photoresponse of the PS surface on different environmental conditions.

II. EXPERIMENTAL

The samples of porous silicon were obtained by anodizing a highly doped n-type (100) silicon with a carrier concentration of $\sim 10^{19}\,\mathrm{cm}^{-3}$ in an aqueous solution of hydrofluoric acid. As a result of anodizing, pores with an average diameter of about 60 nm and an average inter-pore distance of 45 nm were formed on the surface of silicon, which was confirmed by scanning electron microscopy (SEM) (Fig. 1). By changing the current density and electrolyte concentration, it is possible to control the morphology of the resulting PS.

The fabrication of PS and its various morphologies have been described and discussed in detail in previous studies.[35–38] In these references, the morphology of PS was investigated by SEM in both top- and cross-sectional views, revealing dendritic pore-growth. The presence of a native oxide layer has also been reported. The PS samples exhibit a native oxide layer of a few nanometers in thickness, as observed in the Transmission Electron Microscopy (TEM) and corresponding Electron Energy Loss Spectroscopy (EELS) analysis [Figs. 2(a) and 2(b)].

In addition to structural characterization, Raman-spectroscopy was employed to probe structural changes. Raman spectra showed a peak at $522\,\mathrm{cm}^{-1}$ for bare silicon and a slight shift to $520\,\mathrm{cm}^{-1}$ for porous silicon (see Ref. 37) due to stress during the formation process. Fourier Transform Infrared (FTIR) spectroscopy was further used to investigate the interior surface of PS with respect to oxidation (see Ref. 37). FTIR-spectra of as-etched PS-samples exhibited SiH-modes at 2087, 2115, and $2138\,\mathrm{cm}^{-1}$ showing the hydrogen termination of freshly fabricated samples, whereas aged PSi exhibited oxide bands at a wavenumber of about $2250\,\mathrm{cm}^{-1}$ (see Ref. 37).

As mentioned above, our prime experimental probe was the SPV (time-dependent and spectral) performed in different environments, such as N_2 and in high vacuum (10^{-7} Torr). A Besocke Delta Phi GmbH (Jülich, Germany) Kelvin Probe S was positioned near the surface of a sample in a configuration of parallel plates between the reference electrode and the sample. The SPV response was monitored with respect to illumination delivered via a fiber optic bundle, which was coupled through an optical feedthrough to an external optical train. The optical system consisted of a

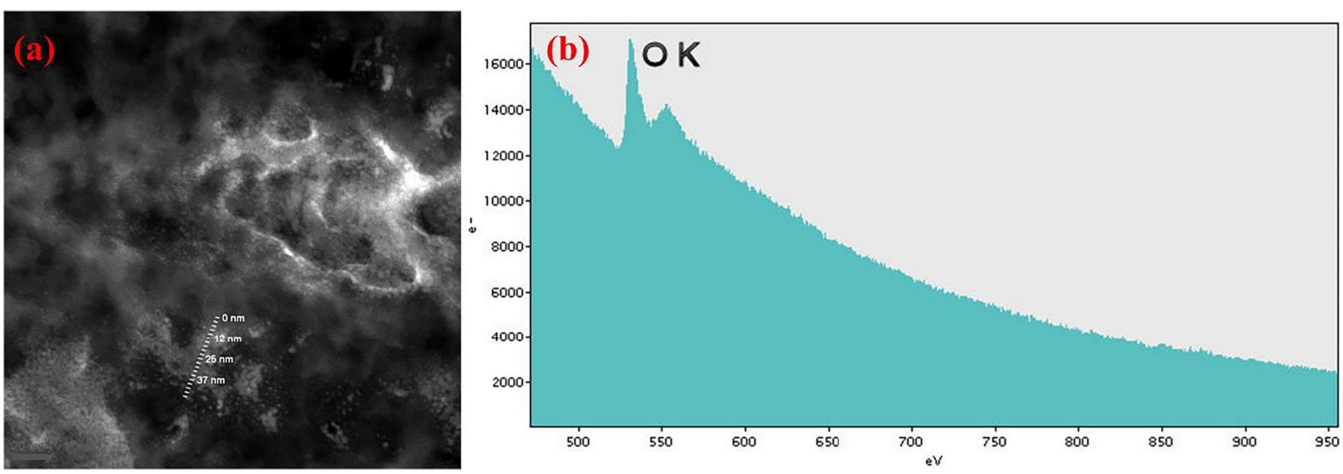

FIG. 2. (a) TEM image of the PS sample showing a native oxide layer, (b) same sample under EELS.

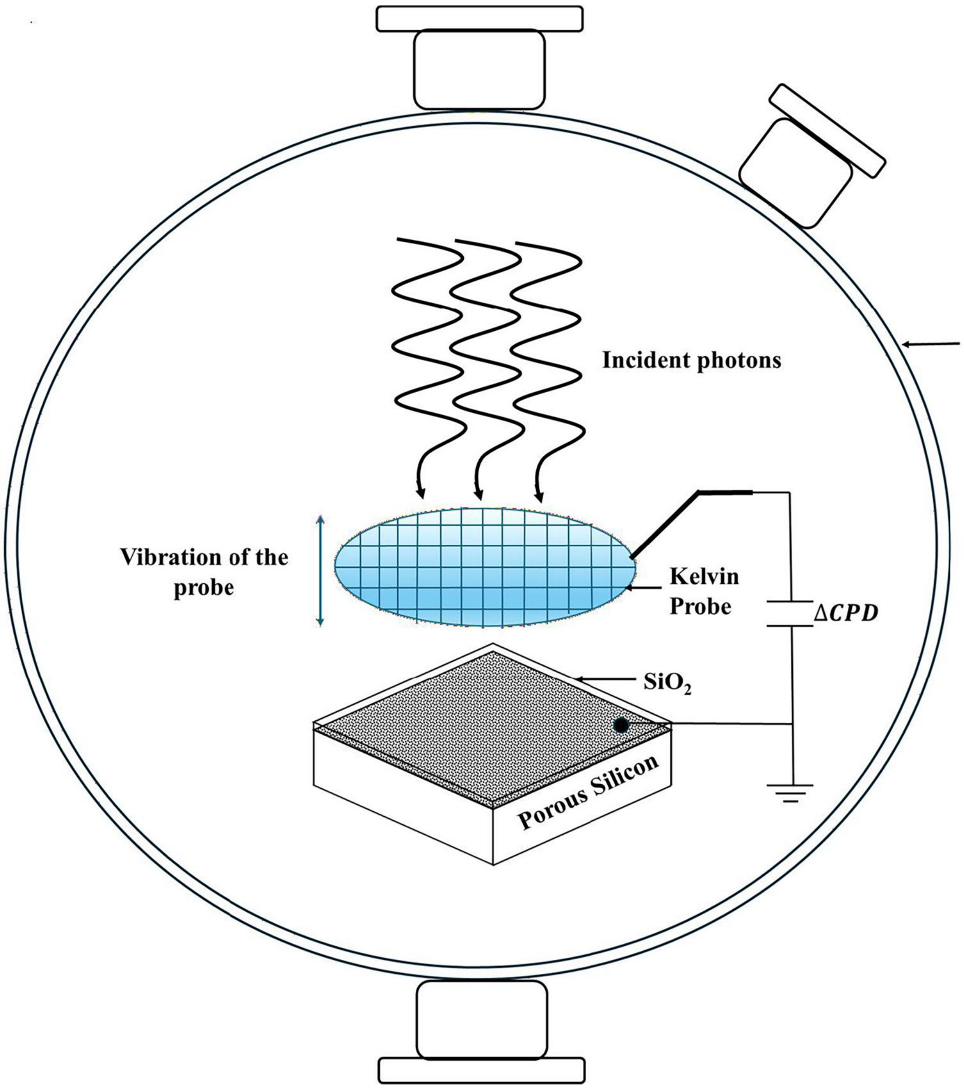

FIG. 3. Schematic illustration of the experimental setup including light source, Kelvin probe, and PS sample for measuring SPV transients under high-vacuum conditions.

250 W quartz tungsten halogen (QTH) lamp as a white light source, a pair of fused silica lenses, a set of bandpass filters, and an Oriel Cornerstone grating monochromator (Newport, Rochester, NY, USA). All SPV measurements were conducted in a high-vacuum chamber (Fig. 3). During the transient SPV measurements, we first recorded a change of the contact potential difference (ΔCPD) between the Kelvin probe and the sample due to the illumination (e.g., with white polychromatic light) allowing the surface to reach equilibrium in light. The surface was then allowed to relax in the dark, and this was again monitored through the CPD reaching saturation. Prior to turning on the illumination, the samples were kept in darkness for an extended time to ensure the initial equilibrium of the surface charge.

III. RESULTS AND DISCUSSION

A. Surface conductivity type of PS

The loss of silicon during the anodization process often causes a decrease in electrical conductivity by several orders of magnitude in the host silicon material. Therefore, for an effective control of the sensing behavior, it is important to know the type of surface conductivity of PS. It has been found that the type of conductivity in porous silicon may differ from that of the original bulk sample.[39] In this regard, one of the key issues in our studies is to determine the type of conductivity of the PS samples under study. As the electronic properties of porous silicon could be very sensitive to environmental gases,[40] the adsorbates may also change

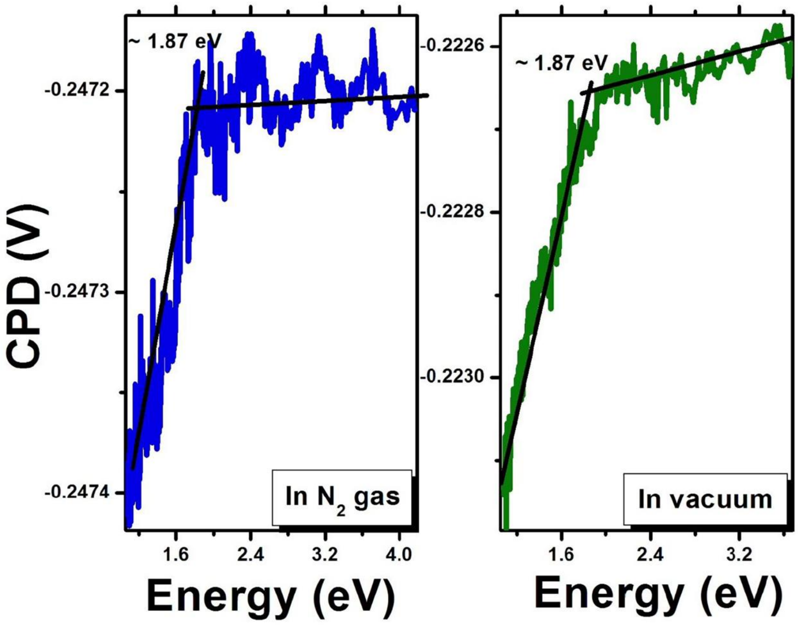

FIG. 4. Bandgap of porous silicon measured in nitrogen and in vacuum by SPV spectroscopy.

the type of surface conductivity. In particular, when trace concentrations of NO_2 and NH_3 are introduced onto a *p*-type surface of PS, a dramatic change is observed, most likely due to hole trapping.[41] In that study, the transient SPV response in the gas ambient revealed a sharp increase and a quick response on the rising front, with a slower component that never reached saturation. During evacuation of gas, a response similar but opposite to that in the case of the gas insertion was observed. This anomalous behavior has not been studied in detail and does not have a satisfactory explanation. In our work, we address such anomalies in detail and incorporate their explanation into our model.

Thus, determining the surface conductivity type of PS is necessary to correctly ascertain the starting assumptions of the model. In our case, these assumptions are based on the initially depleted surface of an *n*-type silicon. To determine the type of surface conductivity, we used the wavelength-dependent SPV measurements.[42] Figure 4 shows a distinct downward transition at ~1.85 eV, presumably due to the PS bandgap broadening. This transition is observed both in vacuum and in nitrogen gas environments, suggesting the *n*-type conductivity of the PS layer. Since the growth of this layer was carried out on an *n*-doped silicon wafer layer, our result means that the conductivity type of the PS layer has not been changed, and the assumptions for the band diagram alignments suggested in this paper are justified.

The origin of the relatively wide bandgap of porous silicon, measured here as well as elsewhere using various optoelectronic methods such as photoluminescence, electroluminescence, and photoabsorption spectroscopy, is still the subject of active debate.[43–46] Usually, two main models are proposed to explain this bandgap broadening phenomenon. The first is based on the radiative recombination of quantum-confined electrons and holes in columnar nanostructures or nanowires that are formed as a result of pore formation. The second model relates the change in the bandgap with the presence of hydrogen or siloxene-like complexes on the surface of porous silicon.[47]

B. Impact of environment on surface charge dynamics

PS is attracting significant attention due to its large surface-to-volume ratio, as well as its high porosity (i.e., the fraction of the volume occupied by pores relative to the total volume of the material). These properties make it a promising material for applications in solar cells,[48] sensors,[49,50] and biological probes.[35] PS often contains a very thin layer of native oxide that forms an additional surface—the SiO_2/Si interface. A significant variety of extrinsic and intrinsic electronic states have been reported in the literature both at the surface of SiO_2 and at the interface between Si and SiO_2 (e.g., Refs. 51 and 52).

An important role in the formation of surface optoelectronic properties is played not only by the interaction of photons with the surface of PS, but also by the chemical environment of this surface. Atoms or molecules of gas from the environment can be physisorbed or chemisorbed, as well as desorbed from the surface of PS. This interaction changes its electronic and optical characteristics. Therefore, it is extremely important to investigate how different gaseous environments affect the charge dynamics of the illuminated surface of the PS.

During irradiation of the PS surface, charge transfer can occur through numerous surface and near-surface states. The rate of this transfer depends on the specific channel and can occur on different time scales. For example, for species adsorbed on the surface, charge transfer may be slower because the bulk charge must overcome a high energy barrier near the surface. Moreover, the pressure of the environment determines the interaction between adsorbed species and PS. At high pressures, surface adsorbates that tend to leave the surface are constantly exposed to collisions with atoms or gas molecules. As a result, these species are brought back to the surface and, as a result, these particles return to the surface and are temporarily physisorbed to the surface. On the other hand, in a high vacuum, the physisorbed atoms or molecules are easily desorbed from the surface during the evacuation process. The processes of adsorption and desorption of foreign species on the surface can significantly change the existing surface states and affect the dynamics of the surface charge. Thus, in this paper, we investigate the effect of ambient pressure on the surface electronic structure of PS, and the surface band bending, in particular, during photostimulation.

Studies of the surface transport properties of PS in different gas environments can help establish performance benchmarks for their potential applications in optoelectronic gas sensors. They could also allow for a deeper understanding of the individual roles played by the Si/SiO_2 interface and the free PS surface in photoinduced charge exchange. In addition, such studies contribute to the elucidation of the reactivity and photochemical properties of the PS surface. Moreover, determining the response and recovery times in light and dark would be beneficial for optimizing the performance of PS-based optoelectronic gas sensors in applications.

For photovoltaic applications of PS, other physical quantities such as excess mobility of charge carriers, absorption coefficient, and material bandgap must be determined. Among these quantities, the key role in determining the efficiency of a solar cell is played by the lifetime of recombination of photogenerated carriers, since it limits the proportion of photo-generated charges collected by the electrodes. The recombination lifetime of photo-excited carriers is defined as the time interval between the absorption of a photon and the recombination of the charge carrier formed by it. For solar cells, as a rule, a long lifetime is desired, which reduces carrier losses and increases the efficiency of the cell.[53] However, the measurement of these times in nanostructured materials such as PS is complicated due to the presence of an indirect bandgap, small film thickness, and its porous structure. Generally, the carrier lifetime is significantly influenced by the presence of interfaces, doping, carrier density, and other factors.

In this regard, the most suitable method for determining the recombination lifetimes and the bandgap in delicate and thin layered materials such as PS is SPV. This technique is non-contact, non-destructive, and highly surface-sensitive.

Previously, only a few studies on this topic have been conducted, separately, either in a vacuum[54] or in air.[55,56] Only Granitzer *et al.*[57] have reported transient surface photovoltage studies of bare and Ni-filled porous silicon performed in different ambients, for extended time. In Ref. 56, a sharp change in surface voltage was recorded when trace amounts of methanol and ammonia were introduced into dry air at normal temperature and pressure. This change was attributed to the photoinduced transitions from and to the surface states caused by the action of gas molecules generated by the gas species. However, this work could not explain the similarities in the observed surface behavior in the dark and under illumination. In another study,[58] SPV measurements on PS on a substrate and in a free-standing form were performed in vacuum using a pulsed laser to study processes occurring on rather short time scales (within a few seconds). Thus, there is a significant deficit of comparative studies between vacuum and various non-vacuum environments, necessitating performing experiments on the PS surface response to different ambient gases.

For the free-standing samples, the barrier height has been found to correlate with porosity. The samples with a low porosity revealed a twofold change in the photoresponse, resulting in both fast and slow processes. On the other hand, in the samples with high porosity, only a slow process was observed. The fast process was attributed to the drift of charge carriers in the region of the space-charge layer, while the slow process was explained by the diffusion of carriers. Another study on the topic in PS showed that excess recombination of carriers at the boundary of the porous layer and the substrate contributes to both fast and slow processes.[55]

Optoelectronic properties of PS, covered with a thin insulating layer, have been actively studied. At the same time, the phototransport properties of such systems have been considered only in a limited number of works. In particular, the surface band bending associated with the slow relaxation times is still not well understood. In samples of oxidized porous silicon, charge capture was detected with characteristic relaxation times of several minutes or more.[54] Both positive and negative surface band bending was observed when irradiating samples with photons of different energies. This opposite nature of band bending was attributed to changes in the electron/hole trapping by surface states after annealing of PS. Studies of photoconductivity[59] and photovoltage[60] of metal/PS junctions in the visible spectral range also revealed multiple relaxation times. For example, changes in barrier height at the junction boundary[61] have been associated with slow relaxation processes. As a rule, metal contacts negatively affect the photoresponse, while the nature of the mechanisms that cause slow relaxation has not yet been fully elucidated, whereas little attention has been paid to the role of the absorbed species remaining at the free surface of PS. In order to fill this gap, in our work, we investigate whether the mechanism of fast photoresponse is related to the intrinsic states and slow response to the extrinsic states.

Herein, we carry out systematic measurements with the variation of multiple experimental parameters in order to elucidate the influence of ambients on the dynamics of light-stimulated surface charge in PS. This allows the formulation of a model that describes the studied phenomena.

SPV transients observed for the pristine PS in different environments are presented in Figs. 5 and 6. These results were obtained first in N_2 gas at a pressure close to atmospheric pressure, and then the experimental chamber was evacuated to 10^{-7} Torr to record transients in vacuum. One can clearly see distinct signal spikes immediately after turning the light on and off both in vacuum and in a nitrogen environment.

These spikes occur on a time scale different from the characteristic time over which the super-bandgap saturation is achieved. In order to compare processes occurring on different time scales, it is convenient to plot the SPV transient curves on a logarithmic time scale. As a result, more informative plots are produced with

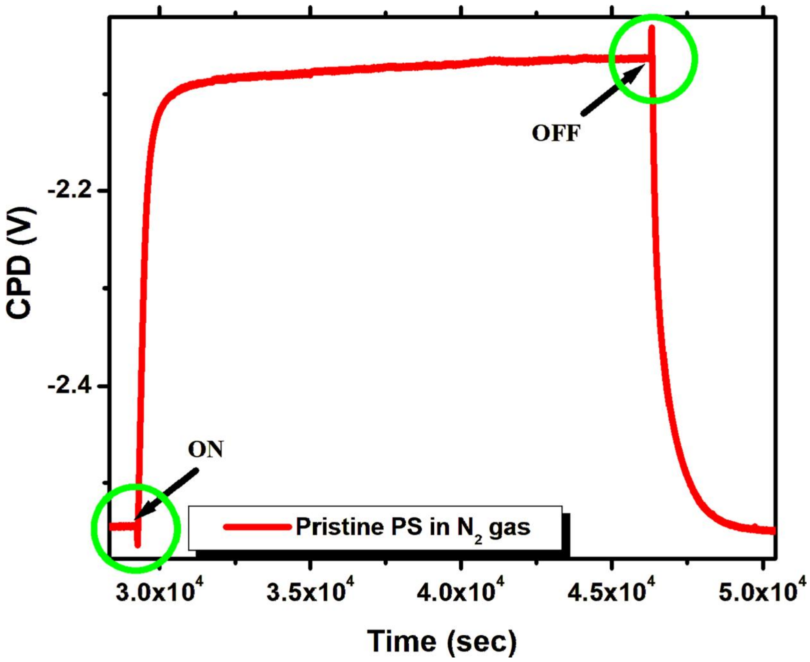

FIG. 5. SPV transient during illumination and in dark on linear time scale for PS in N_2 gas.

the curves revealing both fast and slow components. Therefore, from Fig. 7 –Fig. 10, we use logarithmic time scales. The change in the CPD for the slow process has the opposite direction compared to the change in a fast process. The characteristic time scale for the initial fast phenomena during both the light-on and light-off processes is tens of seconds, while the slower saturation process lasts thousands of seconds. Transients with different timescales—fast and slow—have commonly been observed in a variety of semiconductor materials.[42] A thorough theoretical analysis of multicomponent SPV is provided for wide-band

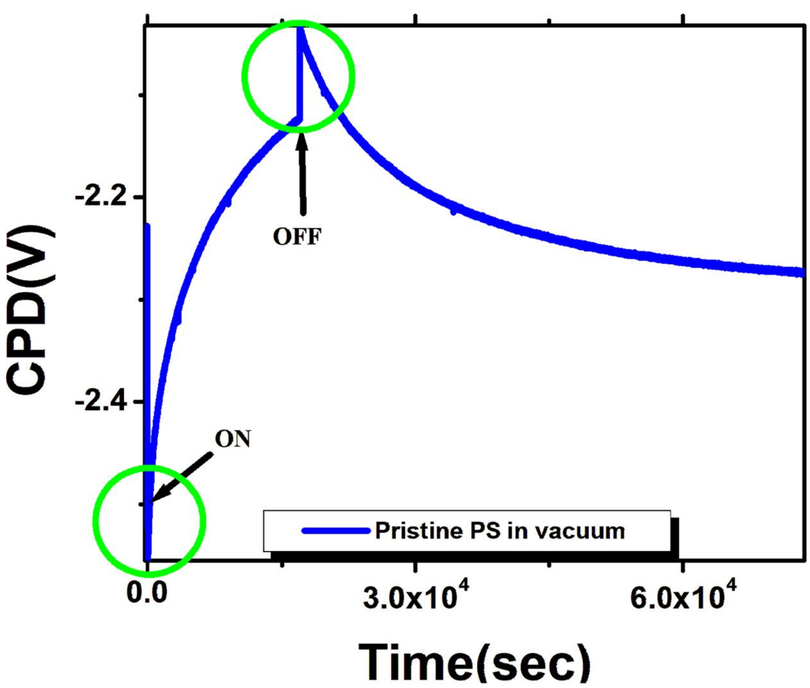

FIG. 6. SPV transient during illumination and in dark on linear time scale for PS in vacuum.

materials such as GaN.[62] According to the authors of this model, the fast and slow components are due, respectively, to internal mechanisms (related to surface or interface states) and external mechanisms (related to the ambient environment). We suggest that this model can also be applied to interpret the phenomena observed in our experiments.

Let us consider a surface band diagram of PS with a thin native oxide layer on the silicon surface as shown in Fig. S1 in the supplementary material. In a gaseous environment, during illumination of PS, the charge transfer is carried out by both internal and external mechanisms, which occur simultaneously, but on different time scales. This leads to the occurrence of multicomponent transients. Since the fast process is observed in both vacuum and gaseous environments, it can be assumed that it is associated with the recombination of charge carriers within PS itself. The slow process in vacuum is weaker than in a gaseous environment, which indicates its possible connection to the physisorbed species already present on the surface of the PS. The fast process during the light-on transient is probably due to the drift of excess electrons in silicon deep into the bulk, as well as the movement of excess holes to the Si/oxide interface under the action of an electric field in the surface space-charge region. The surface recombination of the electrons in the trap states at the interface with the incoming holes from the silicon side can lead to the initial decrease of the CPD in the light-on transient (Fig. S1 in the supplementary material). The decrease in the band bending reduces the depletion width and the barrier height. This allows more electrons to cross the barrier, tunnel through the oxide layer, and become captured by the species at the free surface, possibly converting them into chemisorbed. As a result, the negative charge on the free surface increases, which, in turn, increases the bend bending and leads to a slow increase in the CPD of the light-on transient (Fig. S2 in the supplementary material).

Since chemisorption is a very slow process, it is quite obvious that certain surface-absorbed species contribute to the charge transfer mechanism associated with the slower component. According to Ref. 63, when the surface of PS is exposed to air, it can adsorb molecules such as H_2O, CO_2, H_2, O_2, N_2, F, siloxane complexes, CH_3, etc. The rate of desorption of these species depends on the temperature and pressure of the environment. The difference in the adsorption/desorption rates of those different species present at the free surface may explain the observation of very slow SPV processes.

On the other hand, when similar measurements were made in a vacuum, the slow component became weaker and the SPV signal required very long time to saturate, compared to the case of the N_2 gas environment. We suggest that, in vacuum, a significant amount of the physisorbed species may have been removed from the surface during evacuation. Some of them probably still remain on the surface and may be responsible for trapping the charges from the bulk, thereby contributing to the weaker slow component observed in vacuum during the light-on process.

To test this hypothesis, transient SPV experiments, similar to those described above, were carried out, where the ambient pressure of the nitrogen environment was the variable. Figure 7 shows how the ambient pressure of nitrogen affects the surface band bending during the light-on interval. The pressure varied from atmospheric to high-vacuum conditions, gradually decreasing from experiment to experiment. Before each transient SPV measurement, the pressure in the chamber was stabilized for approximately five hours in the dark, during which the surface potential was monitored. The results obtained showed that the rate of

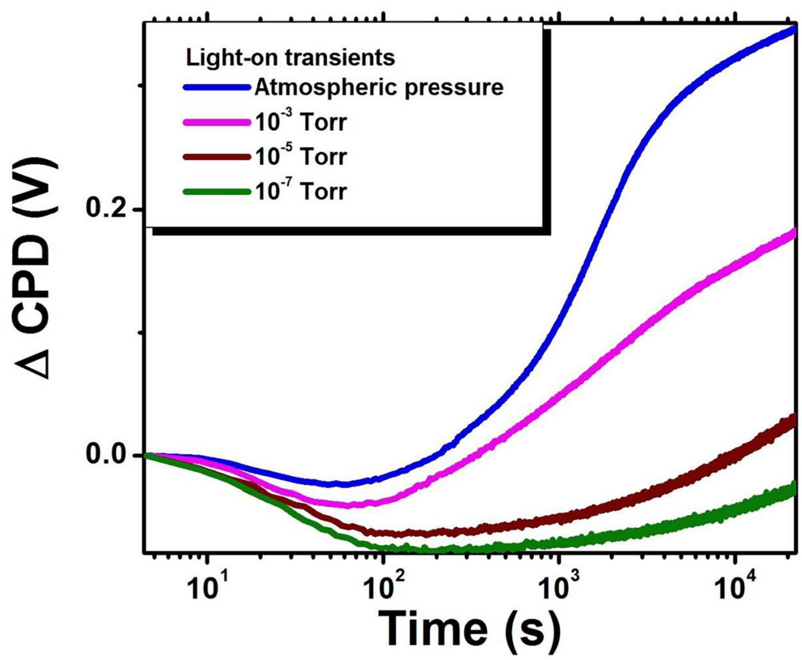

FIG. 7. Pressure-dependent SPV in PS. Slow processes are strongly affected by lowering experimental pressure during illumination.

charge transfer during the fast process was approximately the same for all the ambient pressures. However, for the slow charge recombination processes, significant changes were observed, which is consistent with our hypothesis that the weakening of the slow process with a decrease in pressure is associated with the removal of the inherited adsorbed surface species from the surface. Even at a pressure of 10^{-7} Torr, not all the adsorbed species were completely removed during evacuation. These results credibly confirm the proposed model, according to which the fast charge recombination process is related to the intrinsic states, while the slow process is associated with the surface-adsorbed species.

Additional confirmation of this hypothesis was obtained by studying the reversibility of the observed dependencies. The experiments were carried out with the nitrogen gas pressure cycled from atmospheric through vacuum back to atmospheric. Figure 8 shows the results of this cycling. It is clearly seen that qualitatively the transient behavior is restored when the pressure returns to atmospheric, although a certain hysteresis is observed. It is probably due to the irreversible loss of part of the surface adsorbates during evacuation.

From a practical point of view, such measurements should be useful in assessing the efficiency of this material for applications in optoelectronic gas sensors.

C. Influence of illumination wavelength on photoresponse

Another factor that can affect the surface photoresponse is the photon energy(ies) of the incoming radiation. Surface charge saturation can be achieved with the help of both super- and sub-bandgap excitation. In the case of a sub-bandgap illumination, direct excitation of the surface states takes place, and the efficiency of this process is determined exclusively by the optical cross sections of the corresponding states. As a result, optical processes prevail over thermal ones, which ensures a quick response of the system. On the other hand, in a super-bandgap excitation, the surface states are not excited directly. Their population is modified due to the capture of photoinduced excess charge carriers from the bands. Thus, in this case, the energy positions of the surface states and their thermal cross sections (which control thermal generation/recombination) can compete with optical processes, affecting the overall charge dynamics.[42] As a result, the coexistence of several components in the dynamics of the surface charge can be observed, proceeding on different time scales. Although multicomponent response has already been recorded in materials such as GaN,[64] $SrTiO_3$,[65,66] $LaAlO_3$,[67] GaP,[68] perovskite/Si[69,70] during irradiation with both poly- and monochromatic light separately, no systematic comparison between such excitations for the same material has been carried out so far. PS is no exception in that context. In this work, we investigated the effects of monochromatic super-bandgap and sub-bandgap excitation (as well as their comparison with the case of panchromatic excitation) on the surface band bending in PS for different ambients. These studies provide additional information about the dominant mechanisms of the studied phenomena, since they allow the separation of contributions from intrinsic vs extrinsic electronic states, as well as from the free surface of PS vs the interface with the native oxide.

Most published studies of transient SPV in porous silicon have used a broad-spectrum polychromatic light. Only a few studies have used a monochromatic excitation, and as far as we know, no systematic comparative study using both polychromatic

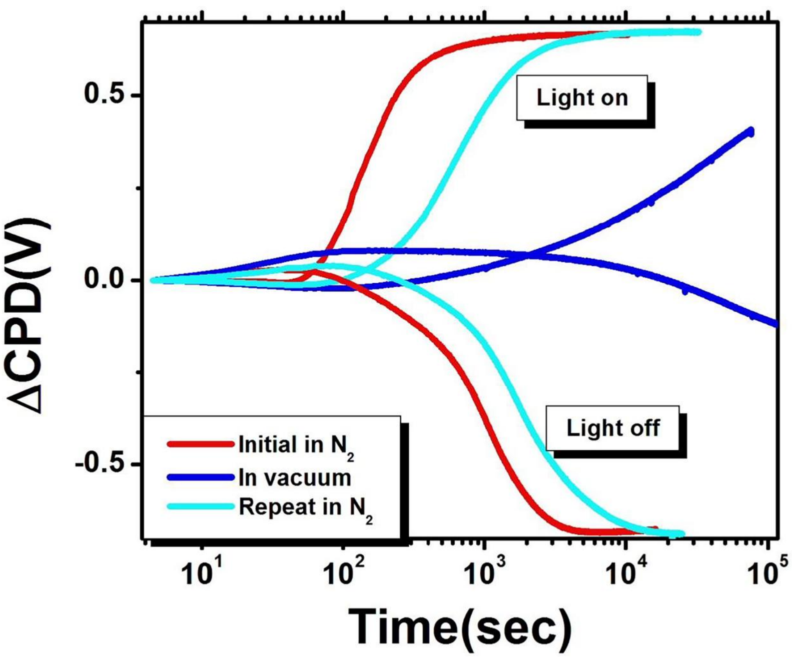

FIG. 8. Light-on and light-off SPV transients in a gas–vacuum–gas cycle.

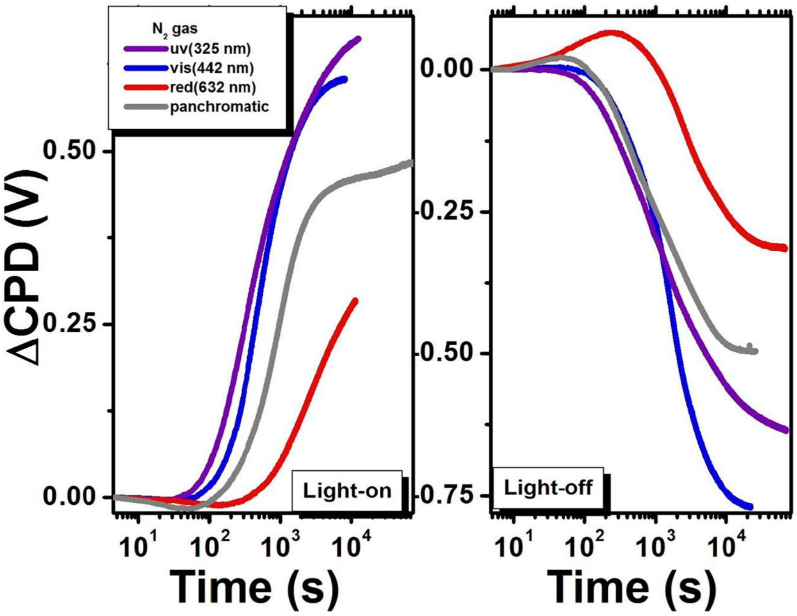

FIG. 9. Light-on and light-off transients during excitation by polychromatic, 325, 442, and 632 nm illumination in N_2.

and monochromatic light has been carried out so far. In our work, we studied SPV transients in both polychromatic and monochromatic light of several different wavelengths in vacuum and in nitrogen gas environments.

Figure 9 juxtaposes the light-on and light-off SPV results employing polychromatic light, ultraviolet (325 nm), blue light (442 nm), and red light (632 nm) in N_2 gas. Figure 10 shows the same set of curves obtained in vacuum.

Qualitatively, in almost all our wavelength-dependent experiments, SPV transients with multiple components were obtained, albeit with different relative contributions. The exception was the near-band edge (632 nm) excitation in vacuum, where only a single SPV transient component was observed. It is possible that, during the near band-edge excitation, only the sub-bandgap charge separation dominates over the thermal excitation of the bulk electrons that reach the surface. The electrons from interface

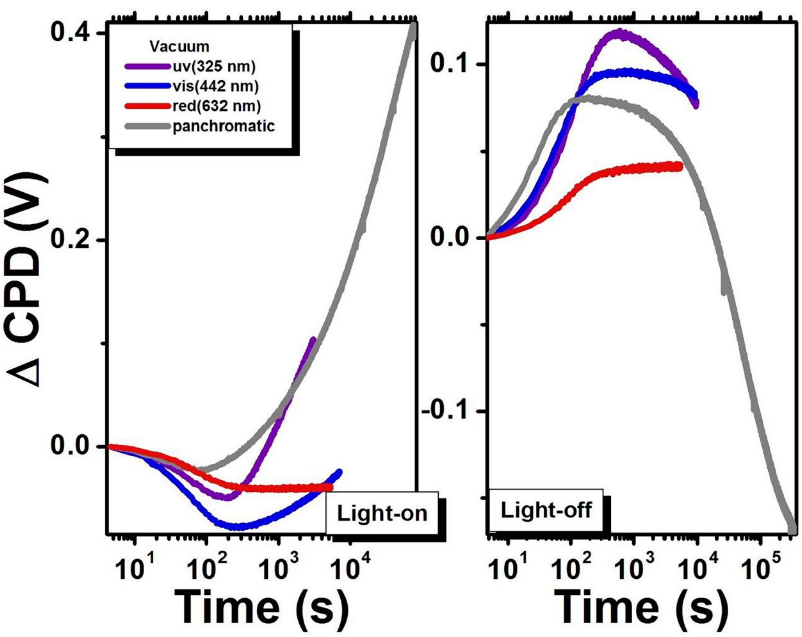

FIG. 10. Light-on and light-off transients excited by polychromatic, 325, 442, and 632 nm illumination in vacuum.

states located near the conduction band at the Si/SiO_2 boundary are promoted to the conduction band and then quickly swept into the bulk due to the strong electric field present at the surface. This, in turn, can lead to a decrease in the band bending and the appearance of only one component in the CPD evolution. For the super-bandgap excitation, the electron–hole pairs are generated in the near-surface region. The flux of holes coming to the surface under the influence of the electric field modifies the electron-rich surface states, while electrons move to the bulk. This gives rise to the fast process. The depletion of the surface electrons reduces the band bending, which, in turn, brings the thermal electrons from the bulk to the surface, yielding the slow process. The relative contribution of the slow process obviously increases with the increasing energy of the incoming photons.

D. Dependence of SPV transients on illumination intensity

Together with the wavelength of excitation, it was found that the intensity of light, which can differ by several orders of magnitude, significantly affects the surface charge dynamics in our samples. The intensity of irradiation has, indeed, a significant impact on various processes that contribute to photoinduced charge separation at a certain wavelength. One way to elucidate this effect is to monitor the initial rate of the surface response to the onset of illumination. The response of the light-induced excess charge carriers is much faster compared to processes induced by thermal emission. Since the initial response of the surface is due primarily to optical excitation processes, it determines the speed of the charge separation in a semiconductor. In the case of common bulk semiconductors, the surface can receive charge from an essentially unlimited reservoir of the bulk region. Whether this applies to nano- or microstructured materials, with a significantly reduced bulk charge reservoir, is a question that requires further investigation.

When analyzing multiple charge recombination processes in bulk semiconductors, the surface is usually considered as the main factor contributing to these processes. Thus, the most likely mechanism of photoinduced charge separation is the drift of excess carriers due to the electric field present in the near-surface region. However, the change in the surface potential barrier can also be affected by a non-uniform absorption, giving rise to the diffusion of carriers, another possible mechanism for charge separation. Both processes—drift and diffusion—can occur simultaneously. Their superposition can, in some cases, reduce the overall efficiency of charge separation.

As in many cases charge dynamics in nanoscale structures differ significantly from those in bulk semiconductors, it is important to investigate how the intensity of illumination affects the initial change of the potential barrier in PS. In previous studies of bulk semiconductors, in particular, GaP,[71] GaN,[64] ZnO,[72] CdS,[28] it has been found that the immediate response of the surface potential change is always linearly dependent on the intensity of illumination. This calls into question whether the linear SPV onset vs light intensity behavior, common for bulk materials, will be retained by essentially nanostructured materials such as PS.

To address this issue, we conducted a series of light intensity-dependent transient SPV experiments on our PS specimens. We investigated how the variation of illumination intensity affects the initial rate of change of the surface band bending in PS.

Our experiments used laser radiation with wavelengths of 325, 442, and 632 nm. The intensity of the incident light was adjusted using neutral density filters. The results observed for the 325 nm excitation are shown in Figs. S3 and S4 in the supplementary material on both linear and logarithmic time scales. One can clearly see the increase in the SPV onset slope with the increasing illumination intensity. Figures S5 and S6 in the supplementary material show SPV transients for the 442 nm excitation with different excitation intensities. Figures S7 and S8 in the supplementary material show SPV transients for the 632 nm excitation with different intensities.

We used the analytical approach proposed by Reshchikov *et al.* for transient SPV results obtained for GaN.[64] In this paper, the authors evaluated the initial slopes of the light-on transients over several decades of photon flux values to determine surface-related parameters. We calculated the onset slopes (i.e., the rate of change of surface potential immediately after turning on/off the light) by using the best linear fit (least squares) for the first five points. The results of these calculations are given in Tables S1–S3 in the supplementary material. The error bars for the slopes were calculated by evaluating the propagation of errors from the measured values of the initial rate of the slope change and the calculated slope. The absolute values of the onset slopes, thus calculated for our samples for the light-on and light-off cycles plotted as a function of illumination intensities, are illustrated in Fig. 11. We found that the slope change during light and dark conditions were approximately the same.

It is important to note that for all the three excitation wavelengths, the onset slopes were not linear functions of the illumination intensity. At the beginning, the values of the slope increase with the increasing intensity, but subsequently reach saturation at higher intensities of laser excitation for wavelengths of 325 and 442 nm. At lower excitation powers, the functional dependencies follow the power law, scaling approximately as the square root of the incident intensity. Such strongly nonlinear behavior, together with saturation at higher powers, may be associated with the limitations on the increase in the excess carrier concentration in PS, which can be explained as follows. In our model, it is assumed that during super-bandgap excitation, electron–hole pairs are generated in the near-surface region as a result of interband transitions, as well as via the modification of interfacial states by holes, due to the band bending being reduced by the field. The corresponding reduction in the surface barrier allows electrons from the bulk to tunnel to the surface and become trapped by the surface adsorbates at the oxide surface. This process would effectively increase the barrier. However, there is a limitation on the change in the current of excess holes with excitation intensity, which is determined by Goodman's model,[42] according to which, electron–hole pairs are generated not only in the SCR but also in the quasi-neutral bulk. The excess holes generated in the SCR drift to the surface due to the electric field, whereas the holes in the quasi-neutral bulk within the diffusion length of the SCR edge, drift *and diffuse* toward the surface. The values of the diffusion

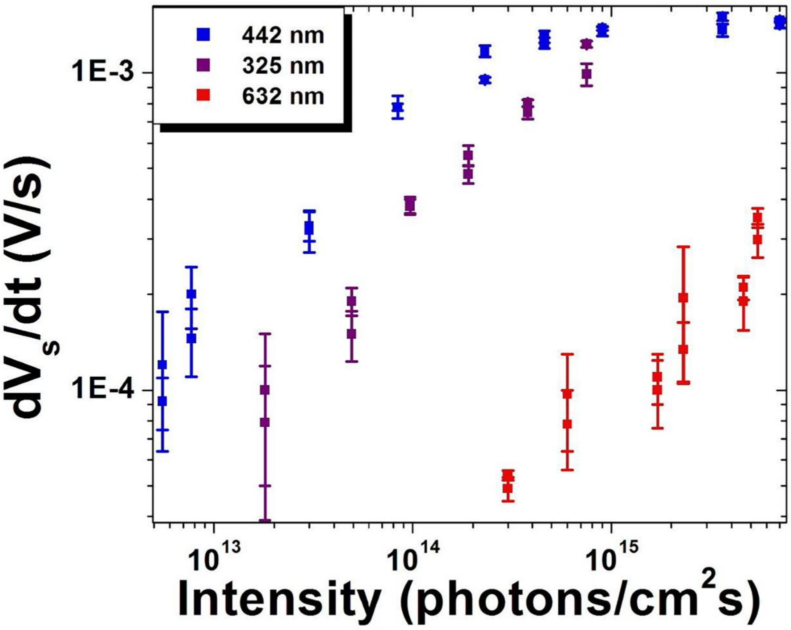

FIG. 11. SPV onset values vs light intensity of incident photons for three different excitation wavelengths.

coefficient for holes D and the diffusion length for holes L are affected by the impurity density. If the drift velocity of holes at the edge of the space-charge region is much smaller than D/L, this model becomes incorrect. Due to the fact that the width of the depletion region is quite small, and the absorption of light decreases almost exponentially from the surface into the bulk, the diffusion from the surface will dominate over the drift. As a result, even for an increasing illumination intensity, the excess holes diffusing toward the bulk will produce saturation of the SPV values.

In addition, it has been reported that depending on the morphology of PS, the mobility of electrons can be two to three orders of magnitude greater than the mobility of holes.[73] The diffusion of the charge carriers can also cause the Dember effect, which affects the initial rate of change of the SPV.[42] Thus, interference of the drift and the Dember effect can cause the nonlinearity in the onset slope vs light intensity dependence. Several additional factors may be involved in limiting the supply of excess carriers, in particular, the ratio of the capture cross sections of holes and electrons, involvement of multiple surface states,[74] as well as interference between super- and sub-bandgap illumination.

IV. CONCLUSIONS

Employing SPV transients, we systematically investigated the influence of the environment on the charge dynamics of PS. Our measurements identified two distinct charge recombination processes that dominate under ambient conditions. The fast process is related to carrier recombination at the Si/SiO_2 interface. As the fast component remains stable regardless of pressure variation, it confirms its intrinsic nature. In contrast, the slow processes are driven by electron trapping via tunneling into surface adsorbates. The significant weakening of this component in a vacuum confirms its dependence on the desorption of surface species. Furthermore, the transition from multi-component to single-component transients under near-band-edge excitation suggests a shift from competing optical and thermal processes to the purely optical depopulation of interface states. Finally, the observed sublinear dependence of the SPV onset slope and its subsequent saturation indicates that carrier transport in PS is governed by the interference between drift and diffusion within the space-charge region.

SUPPLEMENTARY MATERIAL

Please refer to the supplementary material for detailed model band diagrams of the PS surface illustrating fast and slow charge dynamics occurring during illumination under different environments (Figs. S1 and S2). The supplementary material also includes intensity-dependent SPV transients under 325, 442, and 632 nm excitation, presented on both linear and logarithmic time scales (Figs. S3–S8). Corresponding tables (Tables S1–S3) summarizing the onset slopes of SPV transients for respective illumination wavelengths are also provided.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Puskar R. Chapagain: Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Writing – original draft (equal). **Petra Granitzer:** Investigation (equal); Methodology (equal); Validation (equal). **Klemens Rumpf:** Investigation (equal); Methodology (equal); Validation (equal);

Writing – review & editing (equal). **Yuri M. Strzhemechny:** Conceptualization (equal); Methodology (equal); Supervision (equal); Validation (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available within the article and its supplementary material.

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